Benchmarking DAOS Filesystem on Aurora

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DAOS Storage System

Distributed Asynchronous Object Storage (DAOS) is a high performance storage system using key-array-value store allowing end-to-end user space operation, asynchronous I/O operation, and non-blocking I/O.

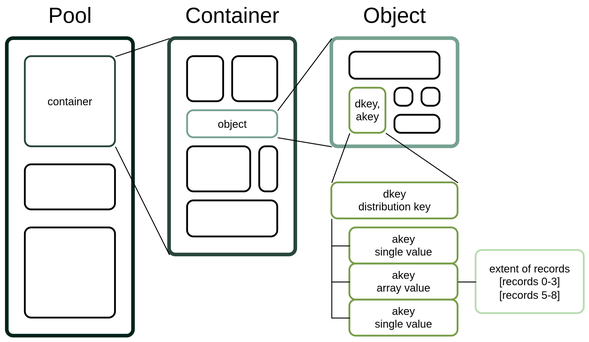

Figure 1. DAOS pool data hierarchy (adapted from [3])

DAOS is provisioned in pools (storage allocations), wherein containers (logical collections of objects) may be created. Objects (key-value stores). Keys have a distribution key (dkey, determining the storage target) and attribute key (akey), with values of either single atomic values or arrays.

DAOS can be accessed via the native libdaos library, libpil4dfs interception library (intercepts I/O and metadata), libioil interception library (intercepts I/O only), and mounting a container via DFuse.

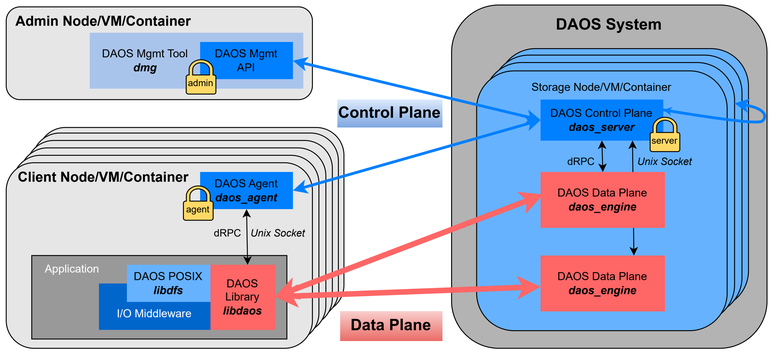

Figure 2. DAOS storage system architecture [2]

Aurora

- Argonne National Lab's Aurora supercomputer has 10,624 compute nodes with 230 PB of DAOS storage on 1024 DAOS server nodes.
- Aurora compute nodes have 8 HPE Slingshot-11 NICs, using dragonfly topology and use two Intel Xeon CPU Max Series, with 52 cores each. For ior we bind to 48 cores per node.
- Our pool had 127 DAOS server nodes (4064 targets, 32 targets per node) with 3 TB Tier 0 (SCM) storage and 97 TB Tier 1 (NVMe) storage.

Acknowledgments

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Benchmarking

We evaluated this storage system using the fio, ior, mdtest, and IO500 I/O benchmarks. We aim to characterize the I/O and metadata performance bottlenecks under a variety of access patterns and client configurations on a pdocutions scale system.

fio Bandwidth

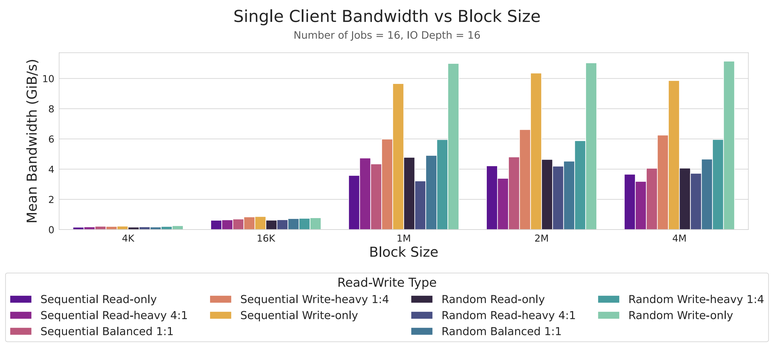

Figure 3. Single client bandwidth vs block size (I/O unit size) for fio using pvsync I/O engine.

- fio bandwidth increased with I/O unit size, saturating at I/O size of 1-2 MiB
- Workloads with more writing, relative to reading, achieved higher throughput via DFuse
- Random vs sequential access had little effect on throughput

IO500

Test	DFS API	POSIX API
ior-easy-write	92.212784 GiB/s	81.309150 GiB/s
ior-easy-read	212.561309 GiB/s	10.761970 GiB/s
ior-hard-write	3.796917 GiB/s	0.665114 GiB/s
ior-hard-read	15.250130 GiB/s	3.159159 GiB/s
Bandwidth Score	32.639612 GiB/s	6.548241 GiB/s

Table 1. 8 node IO500 bandwidth

Test	DFS API	POSIX API
find	775.564385 kIOPS	1.547148 kIOPS
mdtest-easy-write	219.381407 kIOPS	3.927349 kIOPS
mdtest-hard-write	60.646540 kIOPS	1.094255 kIOPS
mdtest-hard-read	807.330148 kIOPS	3.857754 kIOPS
mdtest-easy-stat	1112.188044 kIOPS	4.030321 kIOPS
mdtest-hard-stat	1098.314003 kIOPS	3.849002 kIOPS
mdtest-easy-delete	301.045835 kIOPS	3.389223 kIOPS
mdtest-hard-delete	302.557388 kIOPS	0.386403 kIOPS
IOPS Score	417.712528 kiops	2.185819 kiops

Table 2. 8 node IO500 IOPS

DFS achieved higher bandwidths and IOPS than POSIX on a DFuse mount, especially regarding metadata and reading operations

IOR Latency

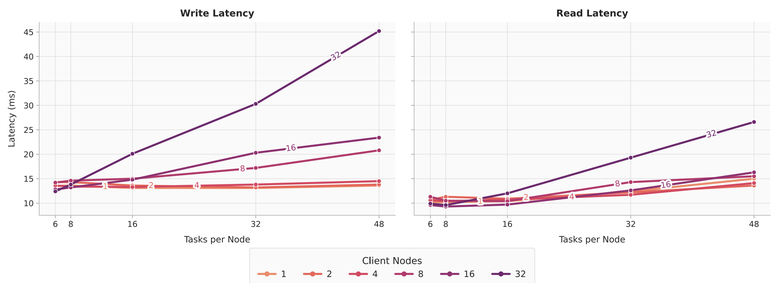

Figure 4. For a transfer size of 1 MiB, the ior latency vs number of tasks, using DFS API.

DFS I/O latency increased with more tasks and nodes, with greater effects on writing than reading

IOR Bandwidth

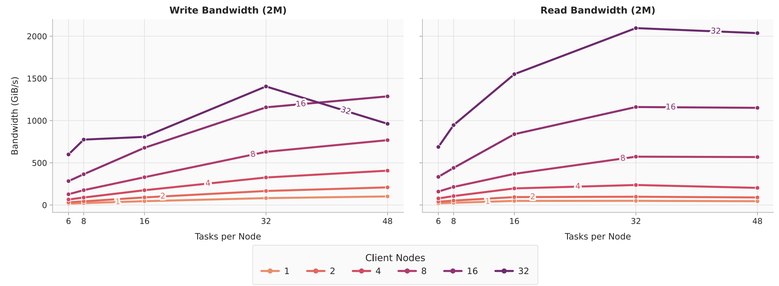

Figure 5. For a transfer size of 2 MiB, the ior bandwidth vs number of tasks, using the DFS API

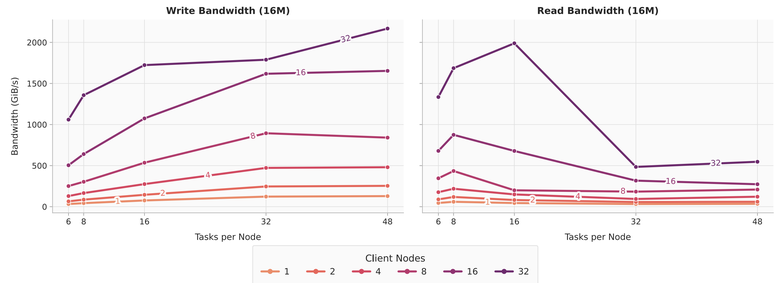

Figure 6. For a transfer size of 16 MiB, the ior bandwidth vs number of tasks, using the DFS API.

- Read and write bandwidth increased with as tasks increased, saturating around 32 tasks
- When a sufficiently large transfer size and number of tasks is used, read bandwith falls

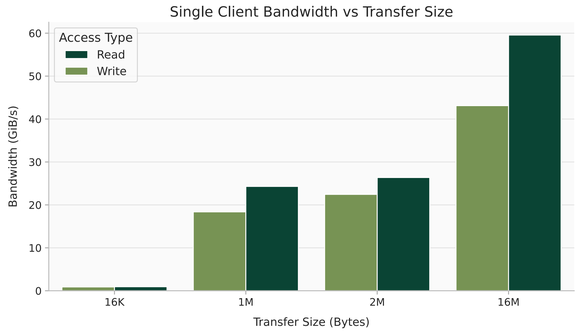

Figure 7. For 8 tasks per node, the ior bandwidth vs number of tasks, using the DFS API.

- For a single client, reading is slightly more performant than writing across transfer sizes
- Sufficiently small transfer sizes (16 KiB) provide low throughput, both for a single client (Figure 7) and all numbers of nodes and tasks per node tested

References

[1] Argonne National Lab. Daos - alcf user guides, 2025. URL `https://docs.alcf.anl.gov/aurora/data-management/daos/daos-overview/`.

[2] Xinxin Mei and et al. Exploring daos as a burst buffer for a 100 gbps daq real-time streaming system. Poster presentation at SC24, November 2024.

[3] DAOS Project. Welcome to daos version 2.6! - daos v2.6, 2025. URL `https://docs.daos.io/latest/`.

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